$3d^3$

July 22, 2026

1 $\left|3d^{\alpha}_{xz}3d^{\alpha}_{xy}3d^{\alpha}_{z^2}3d^{0}_{yz}3d^{0}_{x^2-y^2}\right\rangle$

Here are the CSFs:

$$|1,3/2,3/2\rangle = \left|3d^{\alpha}_{xz}3d^{\alpha}_{yz}3d^{\alpha}_{xy}3d^{0}_{z^2}3d^{0}_{x^2-y^2}\right\rangle$$

$$|2,3/2,3/2\rangle = \left|3d^{\alpha}_{xz}3d^{\alpha}_{yz}3d^{0}_{xy}3d^{\alpha}_{z^2}3d^{0}_{x^2-y^2}\right\rangle$$

$$|3,3/2,3/2\rangle = \left|3d^{\alpha}_{xz}3d^{\alpha}_{yz}3d^{0}_{xy}3d^{0}_{z^2}3d^{\alpha}_{x^2-y^2}\right\rangle$$

$$|4,3/2,3/2\rangle = \left|3d^{\alpha}_{xz}3d^{0}_{yz}3d^{\alpha}_{xy}3d^{\alpha}_{z^2}3d^{0}_{x^2-y^2}\right\rangle$$

$$|5,3/2,3/2\rangle = \left|3d^{\alpha}_{xz}3d^{0}_{yz}3d^{\alpha}_{xy}3d^{0}_{z^2}3d^{\alpha}_{x^2-y^2}\right\rangle$$

$$|6,3/2,3/2\rangle = \left|3d^{\alpha}_{xz}3d^{0}_{yz}3d^{0}_{xy}3d^{\alpha}_{z^2}3d^{\alpha}_{x^2-y^2}\right\rangle$$

$$|7,3/2,3/2\rangle = \left|3d^{0}_{xz}3d^{\alpha}_{yz}3d^{\alpha}_{xy}3d^{\alpha}_{z^2}3d^{0}_{x^2-y^2}\right\rangle$$

$$|8,3/2,3/2\rangle = \left|3d^{0}_{xz}3d^{\alpha}_{yz}3d^{\alpha}_{xy}3d^{0}_{z^2}3d^{\alpha}_{x^2-y^2}\right\rangle$$

$$|9,3/2,3/2\rangle = \left|3d^{0}_{xz}3d^{\alpha}_{yz}3d^{0}_{xy}3d^{\alpha}_{z^2}3d^{\alpha}_{x^2-y^2}\right\rangle$$

$$|10,3/2,3/2\rangle = \left|3d^{0}_{xz}3d^{0}_{yz}3d^{\alpha}_{xy}3d^{\alpha}_{z^2}3d^{\alpha}_{x^2-y^2}\right\rangle$$

$$\begin{aligned}|11,3/2,1/2\rangle =& \frac{1}{\sqrt{3}}\left|3d^{\alpha}_{xz}3d^{\alpha}_{yz}3d^{\beta}_{xy}3d^{0}_{z^2}3d^{0}_{x^2-y^2}\right\rangle + \frac{1}{\sqrt{3}}\left|3d^{\alpha}_{xz}3d^{\beta}_{yz}3d^{\alpha}_{xy}3d^{0}_{z^2}3d^{0}_{x^2-y^2}\right\rangle \\ &+ \frac{1}{\sqrt{3}}\left|3d^{\beta}_{xz}3d^{\alpha}_{yz}3d^{\alpha}_{xy}3d^{0}_{z^2}3d^{0}_{x^2-y^2}\right\rangle\end{aligned}$$

$$\begin{aligned}
|12,3/2,1/2\rangle =&\frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{0}3d_{z^2}^{\beta}3d_{x^2-y^2}^{0}\right\rangle+\frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{0}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{0}\right\rangle\\
&+\frac{1}{\sqrt{3}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{0}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{0}\right\rangle\\
|13,3/2,1/2\rangle =&\frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{0}3d_{z^2}^{0}3d_{x^2-y^2}^{\beta}\right\rangle+\frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{0}3d_{z^2}^{0}3d_{x^2-y^2}^{\alpha}\right\rangle\\
&+\frac{1}{\sqrt{3}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{0}3d_{z^2}^{0}3d_{x^2-y^2}^{\alpha}\right\rangle\\
|14,3/2,1/2\rangle =&\frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{0}\right\rangle+\frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{0}\right\rangle\\
&+\frac{1}{\sqrt{3}}\left|3d_{xz}^{\beta}3d_{yz}^{0}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{0}\right\rangle\\
|15,3/2,1/2\rangle =&\frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{\alpha}3d_{z^2}^{0}3d_{x^2-y^2}^{\beta}\right\rangle+\frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{\beta}3d_{z^2}^{0}3d_{x^2-y^2}^{\alpha}\right\rangle\\
&+\frac{1}{\sqrt{3}}\left|3d_{xz}^{\beta}3d_{yz}^{0}3d_{xy}^{\alpha}3d_{z^2}^{0}3d_{x^2-y^2}^{\alpha}\right\rangle\\
|16,3/2,1/2\rangle =&\frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{0}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle+\frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{0}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle\\
&+\frac{1}{\sqrt{3}}\left|3d_{xz}^{\beta}3d_{yz}^{0}3d_{xy}^{0}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle\\
|17,3/2,1/2\rangle =&\frac{1}{\sqrt{3}}\left|3d_{xz}^{0}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{0}\right\rangle+\frac{1}{\sqrt{3}}\left|3d_{xz}^{0}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{0}\right\rangle\\
&+\frac{1}{\sqrt{3}}\left|3d_{xz}^{0}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{0}\right\rangle\\
|18,3/2,1/2\rangle =&\frac{1}{\sqrt{3}}\left|3d_{xz}^{0}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{0}3d_{x^2-y^2}^{\beta}\right\rangle+\frac{1}{\sqrt{3}}\left|3d_{xz}^{0}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{0}3d_{x^2-y^2}^{\alpha}\right\rangle\\
&+\frac{1}{\sqrt{3}}\left|3d_{xz}^{0}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{0}3d_{x^2-y^2}^{\alpha}\right\rangle\\
|19,3/2,1/2\rangle =&\frac{1}{\sqrt{3}}\left|3d_{xz}^{0}3d_{yz}^{\alpha}3d_{xy}^{0}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle+\frac{1}{\sqrt{3}}\left|3d_{xz}^{0}3d_{yz}^{\alpha}3d_{xy}^{0}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle\\
&+\frac{1}{\sqrt{3}}\left|3d_{xz}^{0}3d_{yz}^{\beta}3d_{xy}^{0}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle\\
|20,3/2,1/2\rangle =&\frac{1}{\sqrt{3}}\left|3d_{xz}^{0}3d_{yz}^{0}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle+\frac{1}{\sqrt{3}}\left|3d_{xz}^{0}3d_{yz}^{0}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle\\
&+\frac{1}{\sqrt{3}}\left|3d_{xz}^{0}3d_{yz}^{0}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle
\end{aligned}$$

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\begin{aligned}
|21,3/2,-1/2\rangle =& \frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{0}3d_{x^2-y^2}^{0}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{0}3d_{x^2-y^2}^{0}\right\rangle \\
&+ \frac{1}{\sqrt{3}}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{0}3d_{x^2-y^2}^{0}\right\rangle \\
|22,3/2,-1/2\rangle =& \frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{0}3d_{z^2}^{\beta}3d_{x^2-y^2}^{0}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{0}3d_{z^2}^{\beta}3d_{x^2-y^2}^{0}\right\rangle \\
&+ \frac{1}{\sqrt{3}}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{0}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{0}\right\rangle \\
|23,3/2,-1/2\rangle =& \frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{0}3d_{z^2}^{0}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{0}3d_{z^2}^{0}3d_{x^2-y^2}^{\beta}\right\rangle \\
&+ \frac{1}{\sqrt{3}}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{0}3d_{z^2}^{0}3d_{x^2-y^2}^{\alpha}\right\rangle \\
|24,3/2,-1/2\rangle =& \frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{0}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{\beta}3d_{yz}^{0}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{0}\right\rangle \\
&+ \frac{1}{\sqrt{3}}\left|3d_{xz}^{\beta}3d_{yz}^{0}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{0}\right\rangle \\
|25,3/2,-1/2\rangle =& \frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{\beta}3d_{z^2}^{0}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{\beta}3d_{yz}^{0}3d_{xy}^{\alpha}3d_{z^2}^{0}3d_{x^2-y^2}^{\beta}\right\rangle \\
&+ \frac{1}{\sqrt{3}}\left|3d_{xz}^{\beta}3d_{yz}^{0}3d_{xy}^{\beta}3d_{z^2}^{0}3d_{x^2-y^2}^{\alpha}\right\rangle \\
|26,3/2,-1/2\rangle =& \frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{0}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{\beta}3d_{yz}^{0}3d_{xy}^{0}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle \\
&+ \frac{1}{\sqrt{3}}\left|3d_{xz}^{\beta}3d_{yz}^{0}3d_{xy}^{0}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle \\
|27,3/2,-1/2\rangle =& \frac{1}{\sqrt{3}}\left|3d_{xz}^{0}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{0}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{0}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{0}\right\rangle \\
&+ \frac{1}{\sqrt{3}}\left|3d_{xz}^{0}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{0}\right\rangle \\
|28,3/2,-1/2\rangle =& \frac{1}{\sqrt{3}}\left|3d_{xz}^{0}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{0}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{0}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{0}3d_{x^2-y^2}^{\beta}\right\rangle \\
&+ \frac{1}{\sqrt{3}}\left|3d_{xz}^{0}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{0}3d_{x^2-y^2}^{\alpha}\right\rangle \\
|29,3/2,-1/2\rangle =& \frac{1}{\sqrt{3}}\left|3d_{xz}^{0}3d_{yz}^{\alpha}3d_{xy}^{0}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{0}3d_{yz}^{\beta}3d_{xy}^{0}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle \\
&+ \frac{1}{\sqrt{3}}\left|3d_{xz}^{0}3d_{yz}^{\beta}3d_{xy}^{0}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle
\end{aligned}
$$

$$|30,3/2,-1/2\rangle = \frac{1}{\sqrt{3}}\left|3d_{xz}^{0}3d_{yz}^{0}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{0}3d_{yz}^{0}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{0}3d_{yz}^{0}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|31,3/2,-3/2\rangle = \left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{0}3d_{x^2-y^2}^{0}\right\rangle$$

$$|32,3/2,-3/2\rangle = \left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{0}3d_{z^2}^{\beta}3d_{x^2-y^2}^{0}\right\rangle$$

$$|33,3/2,-3/2\rangle = \left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{0}3d_{z^2}^{0}3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|34,3/2,-3/2\rangle = \left|3d_{xz}^{\beta}3d_{yz}^{0}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{0}\right\rangle$$

$$|35,3/2,-3/2\rangle = \left|3d_{xz}^{\beta}3d_{yz}^{0}3d_{xy}^{\beta}3d_{z^2}^{0}3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|36,3/2,-3/2\rangle = \left|3d_{xz}^{\beta}3d_{yz}^{0}3d_{xy}^{0}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|37,3/2,-3/2\rangle = \left|3d_{xz}^{0}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{0}\right\rangle$$

$$|38,3/2,-3/2\rangle = \left|3d_{xz}^{0}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{0}3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|39,3/2,-3/2\rangle = \left|3d_{xz}^{0}3d_{yz}^{\beta}3d_{xy}^{0}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|40,3/2,-3/2\rangle = \left|3d_{xz}^{0}3d_{yz}^{0}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|41,1/2,1/2\rangle = +\frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{0}3d_{x^2-y^2}^{0}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{0}3d_{x^2-y^2}^{0}\right\rangle$$

$$|42,1/2,1/2\rangle = +\frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{0}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{0}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{0}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{0}\right\rangle$$

$$|43,1/2,1/2\rangle = +\frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{0}3d_{z^2}^{0}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{0}3d_{z^2}^{0}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|44,1/2,1/2\rangle = +\frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{0}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{0}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{0}\right\rangle$$

$$|45,1/2,1/2\rangle = +\frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{\beta}3d_{z^2}^{0}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{0}3d_{xy}^{\alpha}3d_{z^2}^{0}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|46,1/2,1/2\rangle = +\frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{0}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{0}3d_{xy}^{0}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|47,1/2,1/2\rangle = +\frac{1}{\sqrt{2}}\left|3d_{xz}^{0}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{0}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{0}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{0}\right\rangle$$

$$|48,1/2,1/2\rangle = +\frac{1}{\sqrt{2}}\left|3d_{xz}^{0}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{0}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{0}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{0}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|49,1/2,1/2\rangle = +\frac{1}{\sqrt{2}}\left|3d_{xz}^{0}3d_{yz}^{\alpha}3d_{xy}^{0}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{0}3d_{yz}^{\beta}3d_{xy}^{0}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|50,1/2,1/2\rangle = +\frac{1}{\sqrt{2}}\left|3d_{xz}^{0}3d_{yz}^{0}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{0}3d_{yz}^{0}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|51,1/2,-1/2\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{0}3d_{x^2-y^2}^{0}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{0}3d_{x^2-y^2}^{0}\right\rangle$$

$$|52,1/2,-1/2\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{0}3d_{z^2}^{\beta}3d_{x^2-y^2}^{0}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{0}3d_{z^2}^{\beta}3d_{x^2-y^2}^{0}\right\rangle$$

$$|53,1/2,-1/2\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{0}3d_{z^2}^{0}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{0}3d_{z^2}^{0}3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|54,1/2,-1/2\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{0}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{0}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{0}\right\rangle$$

$$|55,1/2,-1/2\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{\beta}3d_{z^2}^{0}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{0}3d_{xy}^{\alpha}3d_{z^2}^{0}3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|56,1/2,-1/2\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{0}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{0}3d_{xy}^{0}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|57,1/2,-1/2\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{0}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{0}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{0}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{0}\right\rangle$$

$$|58,1/2,-1/2\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{0}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{0}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{0}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{0}3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|59,1/2,-1/2\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{0}3d_{yz}^{\alpha}3d_{xy}^{0}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{0}3d_{yz}^{\beta}3d_{xy}^{0}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|60,1/2,-1/2\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{0}3d_{yz}^{0}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{0}3d_{yz}^{0}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle$$

$$\begin{aligned}|61,1/2,1/2\rangle =& \sqrt{\frac{2}{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{0}3d_{x^2-y^2}^{0}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{0}3d_{x^2-y^2}^{0}\right\rangle \\ &- \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{0}3d_{x^2-y^2}^{0}\right\rangle\end{aligned}$$

$$\begin{aligned}|62,1/2,1/2\rangle =& \sqrt{\frac{2}{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{0}3d_{z^2}^{\beta}3d_{x^2-y^2}^{0}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{0}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{0}\right\rangle \\ &- \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{0}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{0}\right\rangle\end{aligned}$$

$$
\begin{aligned}
|63,1/2,1/2\rangle =& \sqrt{\frac{2}{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{0}3d_{z^2}^{0}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{0}3d_{z^2}^{0}3d_{x^2-y^2}^{\alpha}\right\rangle \\
& - \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{0}3d_{z^2}^{0}3d_{x^2-y^2}^{\alpha}\right\rangle \\
|64,1/2,1/2\rangle =& \sqrt{\frac{2}{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{0}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{0}\right\rangle \\
& - \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{0}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{0}\right\rangle \\
|65,1/2,1/2\rangle =& \sqrt{\frac{2}{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{\alpha}3d_{z^2}^{0}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{\beta}3d_{z^2}^{0}3d_{x^2-y^2}^{\alpha}\right\rangle \\
& - \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{0}3d_{xy}^{\alpha}3d_{z^2}^{0}3d_{x^2-y^2}^{\alpha}\right\rangle \\
|66,1/2,1/2\rangle =& \sqrt{\frac{2}{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{0}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{0}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle \\
& - \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{0}3d_{xy}^{0}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle \\
|67,1/2,1/2\rangle =& \sqrt{\frac{2}{3}}\left|3d_{xz}^{0}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{0}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{0}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{0}\right\rangle \\
& - \frac{1}{\sqrt{6}}\left|3d_{xz}^{0}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{0}\right\rangle \\
|68,1/2,1/2\rangle =& \sqrt{\frac{2}{3}}\left|3d_{xz}^{0}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{0}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{0}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{0}3d_{x^2-y^2}^{\alpha}\right\rangle \\
& - \frac{1}{\sqrt{6}}\left|3d_{xz}^{0}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{0}3d_{x^2-y^2}^{\alpha}\right\rangle \\
|69,1/2,1/2\rangle =& \sqrt{\frac{2}{3}}\left|3d_{xz}^{0}3d_{yz}^{\alpha}3d_{xy}^{0}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{0}3d_{yz}^{\alpha}3d_{xy}^{0}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle \\
& - \frac{1}{\sqrt{6}}\left|3d_{xz}^{0}3d_{yz}^{\beta}3d_{xy}^{0}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle \\
|70,1/2,1/2\rangle =& \sqrt{\frac{2}{3}}\left|3d_{xz}^{0}3d_{yz}^{0}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{0}3d_{yz}^{0}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle \\
& - \frac{1}{\sqrt{6}}\left|3d_{xz}^{0}3d_{yz}^{0}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle \\
|71,1/2,-1/2\rangle =& \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{0}3d_{x^2-y^2}^{0}\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{0}3d_{x^2-y^2}^{0}\right\rangle \\
& - \sqrt{\frac{2}{3}}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{0}3d_{x^2-y^2}^{0}\right\rangle
\end{aligned}
$$

$$|72,1/2,-1/2\rangle = \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{0}3d_{z^2}^{\beta}3d_{x^2-y^2}^{0}\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{0}3d_{z^2}^{\beta}3d_{x^2-y^2}^{0}\right\rangle - \sqrt{\frac{2}{3}}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{0}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{0}\right\rangle$$

$$|73,1/2,-1/2\rangle = \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{0}3d_{z^2}^{0}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{0}3d_{z^2}^{0}3d_{x^2-y^2}^{\beta}\right\rangle - \sqrt{\frac{2}{3}}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{0}3d_{z^2}^{0}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|74,1/2,-1/2\rangle = \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{0}\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{0}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{0}\right\rangle - \sqrt{\frac{2}{3}}\left|3d_{xz}^{\beta}3d_{yz}^{0}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{0}\right\rangle$$

$$|75,1/2,-1/2\rangle = \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{\beta}3d_{z^2}^{0}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{0}3d_{xy}^{\alpha}3d_{z^2}^{0}3d_{x^2-y^2}^{\beta}\right\rangle - \sqrt{\frac{2}{3}}\left|3d_{xz}^{\beta}3d_{yz}^{0}3d_{xy}^{\beta}3d_{z^2}^{0}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|76,1/2,-1/2\rangle = \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{0}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{0}3d_{xy}^{0}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle - \sqrt{\frac{2}{3}}\left|3d_{xz}^{\beta}3d_{yz}^{0}3d_{xy}^{0}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|77,1/2,-1/2\rangle = \frac{1}{\sqrt{6}}\left|3d_{xz}^{0}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{0}\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^{0}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{0}\right\rangle - \sqrt{\frac{2}{3}}\left|3d_{xz}^{0}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{0}\right\rangle$$

$$|78,1/2,-1/2\rangle = \frac{1}{\sqrt{6}}\left|3d_{xz}^{0}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{0}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^{0}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{0}3d_{x^2-y^2}^{\beta}\right\rangle - \sqrt{\frac{2}{3}}\left|3d_{xz}^{0}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{0}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|79,1/2,-1/2\rangle = \frac{1}{\sqrt{6}}\left|3d_{xz}^{0}3d_{yz}^{\alpha}3d_{xy}^{0}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^{0}3d_{yz}^{\beta}3d_{xy}^{0}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle - \sqrt{\frac{2}{3}}\left|3d_{xz}^{0}3d_{yz}^{\beta}3d_{xy}^{0}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|80,1/2,-1/2\rangle = \frac{1}{\sqrt{6}}\left|3d_{xz}^{0}3d_{yz}^{0}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^{0}3d_{yz}^{0}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle - \sqrt{\frac{2}{3}}\left|3d_{xz}^{0}3d_{yz}^{0}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|81,1/2,1/2\rangle = \left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{0}3d_{z^2}^{0}3d_{x^2-y^2}^{0}\right\rangle$$

$$|82,1/2,1/2\rangle = \left|3d_{xz}^{2}3d_{yz}^{0}3d_{xy}^{\alpha}3d_{z^2}^{0}3d_{x^2-y^2}^{0}\right\rangle$$

$$|83,1/2,1/2\rangle = \left|3d_{xz}^{2}3d_{yz}^{0}3d_{xy}^{0}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{0}\right\rangle$$

$$|84,1/2,1/2\rangle = \left|3d_{xz}^{2}3d_{yz}^{0}3d_{xy}^{0}3d_{z^2}^{0}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|85,1/2,1/2\rangle = \left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{0}3d_{z^2}^{0}3d_{x^2-y^2}^{0}\right\rangle$$

$$|86,1/2,1/2\rangle = \left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{2}3d_{z^2}^{0}3d_{x^2-y^2}^{0}\right\rangle$$

$$|87,1/2,1/2\rangle = \left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{0}3d_{z^2}^{2}3d_{x^2-y^2}^{0}\right\rangle$$

$$|88,1/2,1/2\rangle = \left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{0}3d_{z^2}^{0}3d_{x^2-y^2}^{2}\right\rangle$$

$$|89,1/2,1/2\rangle = \left|3d_{xz}^{0}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{0}3d_{x^2-y^2}^{0}\right\rangle$$

$$|90,1/2,1/2\rangle = \left|3d_{xz}^{0}3d_{yz}^{2}3d_{xy}^{0}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{0}\right\rangle$$

$$|91,1/2,1/2\rangle = \left|3d_{xz}^{0}3d_{yz}^{2}3d_{xy}^{0}3d_{z^2}^{0}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|92,1/2,1/2\rangle = \left|3d_{xz}^{0}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{0}3d_{x^2-y^2}^{0}\right\rangle$$

$$|93,1/2,1/2\rangle = \left|3d_{xz}^{0}3d_{yz}^{\alpha}3d_{xy}^{0}3d_{z^2}^{2}3d_{x^2-y^2}^{0}\right\rangle$$

$$|94,1/2,1/2\rangle = \left|3d_{xz}^{0}3d_{yz}^{\alpha}3d_{xy}^{0}3d_{z^2}^{0}3d_{x^2-y^2}^{2}\right\rangle$$

$$|95,1/2,1/2\rangle = \left|3d_{xz}^{0}3d_{yz}^{0}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{0}\right\rangle$$

$$|96,1/2,1/2\rangle = \left|3d_{xz}^{0}3d_{yz}^{0}3d_{xy}^{2}3d_{z^2}^{0}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|97,1/2,1/2\rangle = \left|3d_{xz}^{0}3d_{yz}^{0}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{0}\right\rangle$$

$$|98,1/2,1/2\rangle = \left|3d_{xz}^{0}3d_{yz}^{0}3d_{xy}^{\alpha}3d_{z^2}^{0}3d_{x^2-y^2}^{2}\right\rangle$$

$$|99,1/2,1/2\rangle = \left|3d_{xz}^{0}3d_{yz}^{0}3d_{xy}^{0}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|100,1/2,1/2\rangle = \left|3d_{xz}^{0}3d_{yz}^{0}3d_{xy}^{0}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle$$

$$|101,1/2,-1/2\rangle = \left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{0}3d_{z^2}^{0}3d_{x^2-y^2}^{0}\right\rangle$$

$$|102, 1/2, -1/2\rangle = \left|3d_{xz}^2 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^0 3d_{x^2-y^2}^0\right\rangle$$

$$|103, 1/2, -1/2\rangle = \left|3d_{xz}^2 3d_{yz}^0 3d_{xy}^0 3d_{z^2}^\beta 3d_{x^2-y^2}^0\right\rangle$$

$$|104, 1/2, -1/2\rangle = \left|3d_{xz}^2 3d_{yz}^0 3d_{xy}^0 3d_{z^2}^0 3d_{x^2-y^2}^\beta\right\rangle$$

$$|105, 1/2, -1/2\rangle = \left|3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^0 3d_{x^2-y^2}^0\right\rangle$$

$$|106, 1/2, -1/2\rangle = \left|3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^0\right\rangle$$

$$|107, 1/2, -1/2\rangle = \left|3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^0\right\rangle$$

$$|108, 1/2, -1/2\rangle = \left|3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^0 3d_{z^2}^0 3d_{x^2-y^2}^2\right\rangle$$

$$|109, 1/2, -1/2\rangle = \left|3d_{xz}^0 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^0 3d_{x^2-y^2}^0\right\rangle$$

$$|110, 1/2, -1/2\rangle = \left|3d_{xz}^0 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^\beta 3d_{x^2-y^2}^0\right\rangle$$

$$|111, 1/2, -1/2\rangle = \left|3d_{xz}^0 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^0 3d_{x^2-y^2}^\beta\right\rangle$$

$$|112, 1/2, -1/2\rangle = \left|3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^0\right\rangle$$

$$|113, 1/2, -1/2\rangle = \left|3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^0\right\rangle$$

$$|114, 1/2, -1/2\rangle = \left|3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^0 3d_{x^2-y^2}^2\right\rangle$$

$$|115, 1/2, -1/2\rangle = \left|3d_{xz}^0 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^0\right\rangle$$

$$|116, 1/2, -1/2\rangle = \left|3d_{xz}^0 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^\beta\right\rangle$$

$$|117, 1/2, -1/2\rangle = \left|3d_{xz}^0 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^0\right\rangle$$

$$|118, 1/2, -1/2\rangle = \left|3d_{xz}^0 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^0 3d_{x^2-y^2}^2\right\rangle$$

$$|119, 1/2, -1/2\rangle = \left|3d_{xz}^0 3d_{yz}^0 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^\beta\right\rangle$$

$$|120, 1/2, -1/2\rangle = \left|3d_{xz}^0 3d_{yz}^0 3d_{xy}^0 3d_{z^2}^\beta 3d_{x^2-y^2}^2\right\rangle$$

The Ground State CSFs are 4 14 24 34

$\left\langle 4, \frac{3}{2}, \frac{3}{2} \left| L_x \right| 1, \frac{3}{2}, \frac{3}{2} \right\rangle = -\sqrt{3}I$

Δg_{xx}

$-2\zeta\Delta_{4-1}^{-1}$

$\left\langle 4, \frac{3}{2}, \frac{3}{2} \left| L_y \right| 2, \frac{3}{2}, \frac{3}{2} \right\rangle = I$
$\left\langle 4, \frac{3}{2}, \frac{3}{2} \left| L_y \right| 10, \frac{3}{2}, \frac{3}{2} \right\rangle = -I$

Δg_{yy}

$-\frac{2}{3}\zeta\Delta_{4-2}^{-1}$
$-\frac{2}{3}\zeta\Delta_{4-10}^{-1}$

$\left\langle 4, \frac{3}{2}, \frac{3}{2} \left| L_z \right| 6, \frac{3}{2}, \frac{3}{2} \right\rangle = -2I$
$\left\langle 4, \frac{3}{2}, \frac{3}{2} \left| L_z \right| 7, \frac{3}{2}, \frac{3}{2} \right\rangle = -I$

Δg_{zz}

$-\frac{8}{3}\zeta\Delta_{4-6}^{-1}$
$-\frac{2}{3}\zeta\Delta_{4-7}^{-1}$

For a multiplicity of 4 we have the following:

$D = \frac{1}{2}\left(\left\langle \frac{3}{2}, \frac{3}{2} \middle| \frac{3}{2}, \frac{3}{2} \right\rangle - \left\langle \frac{3}{2}, \frac{1}{2} \middle| \frac{3}{2}, \frac{1}{2} \right\rangle\right)$

$E = \frac{1}{\sqrt{3}}\left\langle \frac{3}{2}, -\frac{1}{2} \middle| \frac{3}{2}, \frac{3}{2} \right\rangle$

$$\begin{aligned}\left\langle \frac{3}{2}, \frac{3}{2} \left| \hat{H}_{eff} \right| \frac{3}{2}, \frac{3}{2} \right\rangle = &- \frac{4}{9}\zeta^2 (\\ &+ \Delta_6^{-1} + \frac{1}{4}\Delta_7^{-1} + \frac{1}{4}\Delta_{11}^{-1} + \frac{1}{12}\Delta_{12}^{-1} + \frac{1}{12}\Delta_{20}^{-1} + \frac{3}{8}\Delta_{41}^{-1} \\ &+ \frac{1}{8}\Delta_{42}^{-1} + \frac{1}{8}\Delta_{61}^{-1} + \frac{1}{24}\Delta_{62}^{-1} + \frac{1}{6}\Delta_{70}^{-1} + \frac{3}{4}\Delta_{82}^{-1} + \frac{1}{4}\Delta_{83}^{-1} \\ &+ \frac{1}{4}\Delta_{95}^{-1} + \frac{3}{4}\Delta_{97}^{-1} \ )\end{aligned}$$

$$
\begin{aligned}
\left\langle \frac{3}{2}, \frac{1}{2} \left| \hat{H}_{eff} \right| \frac{3}{2}, \frac{1}{2} \right\rangle = -\, & \frac{4}{9}\zeta^2 \,(\\
& + \frac{1}{4}\Delta_{1}^{-1} + \frac{1}{12}\Delta_{2}^{-1} + \frac{1}{12}\Delta_{10}^{-1} + \frac{1}{9}\Delta_{16}^{-1} + \frac{1}{36}\Delta_{17}^{-1} + \frac{1}{3}\Delta_{21}^{-1} \\
& + \frac{1}{9}\Delta_{22}^{-1} + \frac{1}{9}\Delta_{30}^{-1} + \frac{1}{6}\Delta_{47}^{-1} + \frac{1}{8}\Delta_{51}^{-1} + \frac{1}{24}\Delta_{52}^{-1} + \frac{8}{9}\Delta_{66}^{-1} \\
& + \frac{1}{18}\Delta_{67}^{-1} + \frac{1}{24}\Delta_{71}^{-1} + \frac{1}{72}\Delta_{72}^{-1} + \frac{1}{18}\Delta_{80}^{-1} + \frac{1}{4}\Delta_{102}^{-1} + \frac{1}{12}\Delta_{103}^{-1} \\
& + \frac{1}{12}\Delta_{115}^{-1} + \frac{1}{4}\Delta_{117}^{-1} \,)
\end{aligned}
$$

$$
\begin{aligned}
\left\langle \frac{3}{2}, -\frac{1}{2} \left| \hat{H}_{eff} \right| \frac{3}{2}, \frac{3}{2} \right\rangle = -\, & \frac{4}{9}\zeta^2 \,(\\
& + \frac{1}{\sqrt{12}}\Delta_{11}^{-1} - \frac{1}{3\sqrt{12}}\Delta_{12}^{-1} - \frac{1}{3\sqrt{12}}\Delta_{20}^{-1} - \frac{\sqrt{3}}{8}\Delta_{41}^{-1} + \frac{\sqrt{3}}{24}\Delta_{42}^{-1} \\
& - \frac{\sqrt{3}}{24}\Delta_{61}^{-1} + \frac{\sqrt{3}}{72}\Delta_{62}^{-1} + \frac{1}{3\sqrt{12}}\Delta_{70}^{-1} + \frac{\sqrt{3}}{4}\Delta_{82}^{-1} - \frac{\sqrt{3}}{12}\Delta_{83}^{-1} \\
& - \frac{\sqrt{3}}{12}\Delta_{95}^{-1} + \frac{\sqrt{3}}{4}\Delta_{97}^{-1} \,)
\end{aligned}
$$

$$
\begin{aligned}
D = & \frac{2}{9}\zeta^2 \,(\\
& + \frac{1}{4}\Delta_{1}^{-1} + \frac{1}{12}\Delta_{2}^{-1} - \Delta_{6}^{-1} - \frac{1}{4}\Delta_{7}^{-1} + \frac{1}{12}\Delta_{10}^{-1} - \frac{1}{4}\Delta_{11}^{-1} \\
& - \frac{1}{12}\Delta_{12}^{-1} + \frac{1}{9}\Delta_{16}^{-1} + \frac{1}{36}\Delta_{17}^{-1} - \frac{1}{12}\Delta_{20}^{-1} + \frac{1}{3}\Delta_{21}^{-1} + \frac{1}{9}\Delta_{22}^{-1} \\
& + \frac{1}{9}\Delta_{30}^{-1} - \frac{3}{8}\Delta_{41}^{-1} - \frac{1}{8}\Delta_{42}^{-1} + \frac{1}{6}\Delta_{47}^{-1} + \frac{1}{8}\Delta_{51}^{-1} + \frac{1}{24}\Delta_{52}^{-1} \\
& - \frac{1}{8}\Delta_{61}^{-1} - \frac{1}{24}\Delta_{62}^{-1} + \frac{8}{9}\Delta_{66}^{-1} + \frac{1}{18}\Delta_{67}^{-1} - \frac{1}{6}\Delta_{70}^{-1} + \frac{1}{24}\Delta_{71}^{-1} \\
& + \frac{1}{72}\Delta_{72}^{-1} + \frac{1}{18}\Delta_{80}^{-1} - \frac{3}{4}\Delta_{82}^{-1} - \frac{1}{4}\Delta_{83}^{-1} - \frac{1}{4}\Delta_{95}^{-1} - \frac{3}{4}\Delta_{97}^{-1} \\
& + \frac{1}{4}\Delta_{102}^{-1} + \frac{1}{12}\Delta_{103}^{-1} + \frac{1}{12}\Delta_{115}^{-1} + \frac{1}{4}\Delta_{117}^{-1} \,)
\end{aligned}
$$

$$
\begin{aligned}
E =& \frac{4}{9\sqrt{3}}\zeta^2 \, (\\
& - \frac{1}{\sqrt{12}}\Delta_{11}^{-1} + \frac{1}{3\sqrt{12}}\Delta_{12}^{-1} + \frac{1}{3\sqrt{12}}\Delta_{20}^{-1} + \frac{\sqrt{3}}{8}\Delta_{41}^{-1} - \frac{\sqrt{3}}{24}\Delta_{42}^{-1} \\
& + \frac{\sqrt{3}}{24}\Delta_{61}^{-1} - \frac{\sqrt{3}}{72}\Delta_{62}^{-1} - \frac{1}{3\sqrt{12}}\Delta_{70}^{-1} - \frac{\sqrt{3}}{4}\Delta_{82}^{-1} + \frac{\sqrt{3}}{12}\Delta_{83}^{-1} \\
& + \frac{\sqrt{3}}{12}\Delta_{95}^{-1} - \frac{\sqrt{3}}{4}\Delta_{97}^{-1} \,)
\end{aligned}
$$